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Advances in Space Research 58 (2016) 145–154

ADVANCES IN SPACE RESEARCH (a COSPAR publication)

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Vibratory compaction method for preparing lunar regolith drilling simulant

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Received 24 December 2015; received in revised form 22 April 2016; accepted 24 April 2016
Available online 29 April 2016

Abstract

Drilling and coring is an effective way to acquire lunar regolith samples along the depth direction. To facilitate the modeling and simulation of lunar drilling, ground verification experiments for drilling and coring should be performed using lunar regolith simulant. The simulant should mimic actual lunar regolith, and the distribution of its mechanical properties should vary along the longitudinal direction. Furthermore, an appropriate preparation method is required to ensure that the simulant has consistent mechanical properties so that the experimental results can be repeatable. Vibratory compaction actively changes the relative density of a raw material, making it suitable for building a multilayered drilling simulant. It is necessary to determine the relation between the preparation parameters and the expected mechanical properties of the drilling simulant. A vibratory compaction model based on the ideal elastoplastic theory is built to represent the dynamical properties of the simulant during compaction. Preparation experiments indicated that the preparation method can be used to obtain drilling simulant with the desired mechanical property distribution along the depth direction.
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Keywords: Lunar regolith; Drilling simulant; Simulant preparation; Vibratory compaction

1. Introduction

Lunar regolith refers to the 4–15 m thick layer of granular materials covering the underlying bedrock of the moon (Heiken et al., 1991). Drilling and coring is known to be an effective way to acquire regolith samples while maintaining their stratification along the depth direction (Bar-Cohen and Zacny, 2009). The Chang'e-5 lunar exploration project aims to sample lunar regolith through an automated drilling and coring device and return it to Earth. The drill and lunar regolith have a complex physical interaction that may determine whether the device can obtain a sample. To verify the designed drill auger and bit and whether the drilling strategy is appropriate, ground experiments should be conducted with the desired drilling simulant (Bar-Cohen and Zacny, 2009; Gouache et al., 2011; Pitcher and Gao, 2015). Therefore, an appropriate drilling simulant must be prepared to ensure that the distribution of the mechanical properties is reproducible along the longitudinal direction.

The preparation of a regolith simulant is a prerequisite for testing the performance of a space-based device that will interact with the regolith. Landers, lunar surface excavation scoops, and rovers make contact with only a shallow layer of regolith, the mechanical properties of which can be considered consistent at one specific site (Zhong, 2012; Agui et al., 2013; Li et al., 2015). However, because the mechanical properties of lunar regolith vary with the landing site, homogenous regolith simulants with different mechanical properties may be preferable for tests of landing, surface regolith excavation, and sinkage mobility. To acquire subsurface lunar regolith samples, drilling devices must

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http://dx.doi.org/10.1016/j.asr.2016.04.021
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make contact with both a shallow and a deep layer of regolith. Generally, lunar regolith has low to medium relative density in the shallow layer and high relative density just 10–20 cm below the lunar surface (Mitchell et al., 1974). To verify the design of a drilling device, the drilling simulant should be prepared to mimic actual lunar regolith with varying distribution of mechanical properties along the depth direction; a simulant with high relative density is indispensable for ground drilling experiments. Pour, rain, vibratory compaction, Proctor compaction, and static compaction methods were used to prepare homogeneous regolith simulants with different relative densities (Gouache et al., 2010; Zhong, 2012; Scott and Saaj, 2009). The pour and rain methods were used to prepare loose or medium-density simulants, in which the relative density was governed by the falling height and flow rates (Gouache et al., 2010; Jiang et al., 2013). The vibratory compaction method was used to prepare medium- or high-density simulants (Klosky et al., 2000). In this method, the relative density could be controlled by changing the vibratory parameters (e.g., vibratory frequency and impact force) (Wersäll and Larsson, 2013). The Proctor compaction method was mainly used for measuring the mechanical properties of standard simulant specimens (Zhong, 2012; MWR, 1999). The tamping method was used to prepare large-scale soil samples (Kleinhenz and Wilkinson, 2012; Adam et al., 2007). The static compaction method was used to compact a simulant by using compression force (Scott and Saaj, 2009). Preparation using the static compaction method is time-consuming when producing a homogenous soil specimen, especially when the raw materials have weak flow characteristics (Zheng et al., 2008). While numerous methods have been used to prepare homogenous simulants, few studies have reported on the preparation of drilling simulants with mechanical properties that vary along the longitudinal direction. Because vibratory compaction can actively change the simulant density under dynamic loads, it is considered appropriate for producing a multilayered drilling simulant. Therefore, in this study, vibratory compaction was used to prepare the lunar regolith drilling simulant to simulate the subsurface lunar regolith.

The preparation of the regolith drilling simulant can be facilitated by understanding the effect of vibratory compaction on the simulant. The finite element method (FEM) and discrete element method (DEM) have been used to describe the mechanical behavior of granular materials during vibratory compaction (Kenneally et al., 2015; Masad et al., 2010; Arnold and Herle, 2009; Chen et al., 2013). However, FEM cannot well model large deformation because it suffers from mesh distortion, which may result in low calculation accuracy and convergence failure. Although DEM simulation can overcome this limitation, the high computation cost restricts DEM application to small-scale or short-duration simulation. Analytical models have also been used to describe the behavior of granular materials during vibratory compaction. Most analytical models such as the Maxwell, Kelvin, Burgers, and Huet–Sayegh models represented granular materials as a combination of springs and dampers (Sysante and Mooney, 2008; Kordestani et al., 2010; Beainy et al., 2014; Xu and Solaimanian, 2009). Because the damping parameters of soil vary during vibratory compaction and depend on the moisture content, density, and type of soil, the damping parameters are difficult to determine and validate during compaction (Beainy et al., 2014; Facas et al., 2010). The ideal elastoplastic theory can be used to describe the behavior of soils under dynamic loads, and the parameters of the model based on this theory are convenient to determine and verify (Niu, 2003). However, the ideal elastoplastic theory has not yet been adopted to describe soil behavior during vibratory compaction.

This paper presents a vibratory-compaction-based method to prepare lunar regolith drilling simulant for ground drilling and coring experiments. Based on the ideal elastoplastic theory, a vibratory compaction model is built to represent the behavior of lunar regolith simulant during vibratory compaction in simulant granular systems. Experiments are conducted using a vibratory table to validate the vibratory compaction model. Inspired by the Apollo analysis method, a relation between the preparation parameters and the expected mechanical properties of the drilling simulant is developed to facilitate simulant preparation. Terrestrial preparation experiments are conducted to obtain a lunar regolith drilling simulant with desired mechanical property distribution along the depth direction.

The remainder of this paper is organized as follows. First, a review of lunar regolith is presented. Then, the lunar regolith simulant material is introduced. The vibratory compaction model is built to describe the dynamic behavior of the lunar regolith simulant during compaction. Nextthe vibratory compaction model is experimentally validated. Finally, preparation experiments are conducted to obtain a lunar regolith drilling simulant.

2. Review of lunar regolith

To analyze the lunar regolith acquired by the Apollo missions, the Apollo method was developed to analyze the mechanical property distribution of lunar regolith along the longitudinal direction (Heiken et al., 1991). The core samples collected by the Apollo Lunar Surface Drill and the core tubes were used to analyze the relation between the depth and the density of lunar regolith (Bar-Cohen and Zacny, 2009). Ground physical tests based on the basaltic lunar regolith simulant yielded the relation between the shear strength and the relative density (Mitchell et al., 1972). The results of these two analyses indicate that the mechanical property distribution of lunar regolith can be acquired along the depth direction.

According to the Apollo analysis results, the relative density of lunar regolith is generally low to medium in the shallow layer and high just 10–20 cm below the lunar surface (Mitchell et al., 1974). The fitting equation of density versus depth can be expressed as a logarithmic function:

$$\rho = \rho_0 + k\ln(z+1) \tag{1}$$

where ρ_0 is the density of the shallow layer regolith; k, a constant value; and z, the sampling depth. The lunar regolith density increases rapidly with depth in the shallow layer and approaches a stable value in the deep layer. This density distribution may be caused by meteoritic impacts and vibrations.

The shear strength that influences the drilling load and coring rate in the drilling process must be mimicked in regolith simulant preparation. However, studies have not reported in situ test data of the lunar regolith shear strength along the depth direction. Ground simulation tests were conducted to determine the shear strength of the lunar regolith with different relative densities (Carrier, 1972). Based on the basaltic simulant test data (Mitchell et al., 1972, 1974), the fitting equation of the relation between the shear strength and the relative density can be expressed as follows:

$$\begin{cases} \tan\phi = \frac{1.256}{1.7-D_r} - 0.263 \\ \lg c = -9.974\frac{1.7-D_r}{2.7-D_r} + 4.839 \end{cases} \tag{2}$$

where ϕ is the internal friction angle; c, the cohesion; and D_r, the relative density.

According to the Apollo estimation of the regolith density distribution and the relation of the shear strength and the density, the shear strength distribution of lunar regolith can be acquired along the depth direction. Fig. 1 shows the Apollo method for analyzing the in situ lunar regolith mechanical property distribution along the depth direction in the Apollo missions, and Eq. (3) is the definition of the porosity n, void ratio e, density ρ, specific gravity G and relative density D_r. In this method, the key parameter is the density that can be accounted for directly by the Apollo core samples. The lunar regolith shear strength is related to the density, and it was acquired by analogy with the shear strength of the basaltic simulant (according to Eqs. (2) and (3)). Though the mechanical properties of the basaltic simulant bracket that of the actual lunar regolith, the compositions of the basaltic simulant are not exactly the same as those of the lunar regolith. The Apollo analysis method may allow conversions between the mechanical properties of lunar regolith with limited accuracy. Despite the limitation of the Apollo method, this method provides a way to represent all mechanical properties by a single parameter and can be employed in the preparation and measurement of lunar regolith drilling simulant.

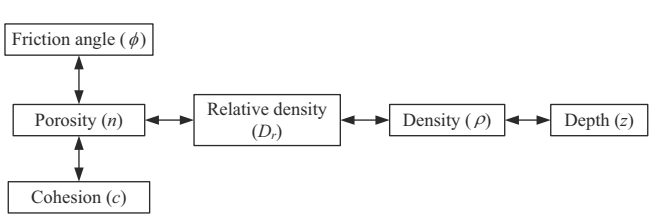

Fig. 1. Apollo analysis method.

$$\begin{cases} D_r = \frac{e_{max}-e}{e_{max}-e_{min}} \\ e = \frac{n}{1-n} \\ \rho = G\rho_w(1-n) \end{cases} \tag{3}$$

where ρ_w is the water density; e_{max}, the maximum void ratio; and e_{min}, the minimum void ratio.

3. Lunar regolith simulant material

The lunar regolith simulant used in this study was made of basaltic pozzolana collected from Nanjing, China. Table 1 shows the comparison of the mineral compositions of the pozzolana and the lunar regolith from the Apollo 14 landing site (Laul et al., 1982).

The raw materials were ground to different particle sizes by a Raymond mill. According to the grain-size distribution of the actual lunar regolith (Carrier, 1973), the particle size range of the simulant is confined to 0.1–1 mm, and the median grain size is 0.41 mm. The shapes of the simulant particles are mainly angular, subangular, and elongated, as shown in Fig. 2 (Shi, 2015). Physical property tests (following the SL237-1999 standard procedure (MWR, 1999)) revealed that the minimum density, maximum density, specific gravity, minimum void ratio, and maximum void ratio were 1.63 g/cm^3, 2.15 g/cm^3, 2.94 g/cm^3, 0.37, and 0.78, respectively.

3.1. Compressibility

The shapes of the simulant particles vary greatly, ranging from spherical to angular, and the surfaces of the particles are coarse. The interlocking among angular simulant particles is dramatic in simulant samples, and inter-particle sliding among angular particles is more difficult than that among smooth particles (Shinohara et al., 2000; Li, 2013; Zhao et al., 2015). When samples of the raw simulant materials are prepared, the solid skeleton is more stable than soils with smooth particles. The simulant particle shapes strongly influence the compressibility and shear strength of the simulant. It is essential to understand the compressibility and shear strength of the simulant samples before preparing the simulant samples and analyzing the drilling and coring processes.

Confined compression tests were performed to determine the simulant compression properties. Fig. 3 shows the confined compression behavior of the simulant. As

Table 1
Comparison of mineral compositions of pozzolana and lunar regolith.

Component	Proportions (%)	
	Pozzolana	Apollo 14
Plagioclase	59.4	58–69
Olivine	16.5	2–10
Pyroxene	14.1	0–3
Opaque mineral	4.5	0
Volcanic glass	5	10–20

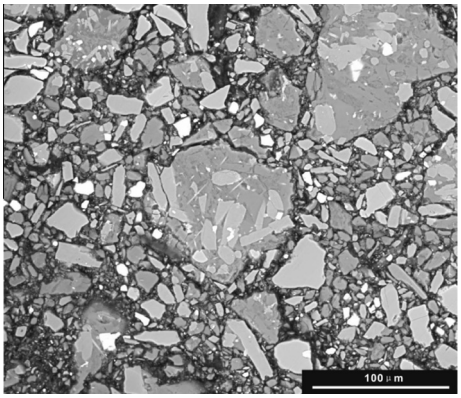

Fig. 2. Photomicrograph of lunar regolith simulant particles.

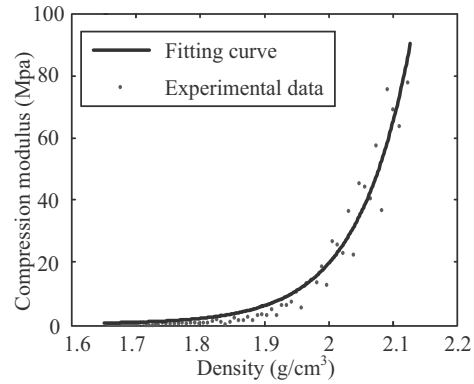

Fig. 4. Relation between compression modulus and density of the simulant.

the simulant sample is loose, indicating that the sample contains large voids, small particles may easily fill the voids among large particles. Therefore, the simulant sample volume decreased rapidly when the simulant density was low. The probability of contact among particles increases with a decrease in the sample voids, which may restrain interparticle sliding. As the sample becomes dense, the solid skeleton approaches the stable state, and the compressibility of the simulant decreases.

3.1.1. Compression modulus

Fig. 4 shows the relation between the compression modulus and the density of the simulant, which is determined by conducting a confined compression test. The test results can be fitted by Eq. (4). The compression modulus increases smoothly when the density of the simulant is lower than 1.95 g/cm^3. The solid skeleton of the simulant becomes more stable as the density increases, and thus, the simulant compression modulus increases significantly when its density is higher than 2 g/cm^3.

$$E = 6.498 \times 10^{-10} \exp(12.07\rho) \tag{4}$$

where ρ is the density.

3.1.2. Resilient modulus

Laboratory bearing plate tests were conducted to measure the resilient modulus of the simulant. Unconfined resilient moduli were obtained as test results. To use these results to determine the model parameters mentioned in Section 3, the unconfined resilient modulus was converted to a confined one. The ratios of compression deformation to resilient deformation (RCRD) measured by the tests were used to build the relation between the confined resilient modulus and the compression modulus:

$$E_{su} = k_{cr}E \tag{5}$$

where $k_{cr} = l_r/l$, l_r is the resilient deformation, and l is the compression deformation.

Fig. 5 shows the relation between the RCRD and the density of the simulant. The test results can be fitted by the following equation:

$$k_{cr} = 5.686 \times 10^{12}\rho^{-41.58} + 0.9079 \tag{6}$$

3.2. Shear strength

Direct shear tests were used to determine the shear strength of the simulant samples at different relative densities. The tests were conducted under pressures of 100, 200, 300, and 400 kPa to determine the shear strength of the simulant samples at relative densities of 75%, 80%, 85%, 90%, 95%, and 100%. By using the Mohr–Coulomb law, the internal friction angle ϕ and apparent cohesion c of the simulant were determined for each sample. The test

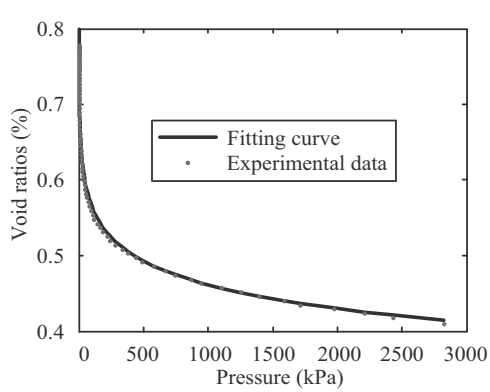

Fig. 3. Relation between void ratio and compression pressure.

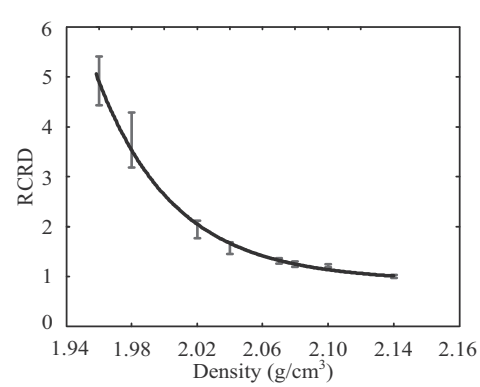

Fig. 5. Ratios of compressive and resilient deformation of the simulant.

Table 2
Shear strength of the lunar regolith simulant.

Relative density (%)	75	80	85	90	95	100
Angle of internal friction (°)	30.5	31.4	32.3	33.3	34.2	35.6
Apparent cohesion (kPa)	0.33	0.93	1.47	2.08	2.72	2.80

results revealed a trend of higher c and higher ϕ for the simulant with higher density, as shown in Table 2, and these results can be fitted as follows:

$$\begin{cases} \phi = & \arctan(0.499D_r + 0.2109) \\ c = & 0.01207\exp(5.558D_r) \end{cases} \tag{7}$$

4. Vibratory-compaction-based preparation

To simulate the lunar regolith at different landing sites or in deep layers, simulant materials for preparing simulant samples with different relative densities are required. The vibratory compaction method described above can actively change the simulant density under dynamic loads and is therefore appropriate for preparing a multilayered drilling simulant. A vibratory table (see Fig. 6) was used to prepare the simulant samples. A vibratory motion was produced using a rotating eccentric mass, inducing vertical base motions of 1 mm. The vibratory compaction process can be simplified to the interaction of a mass and the simulant. This section presents a vibratory compaction model to describe the interaction of the mass and the simulant during vibratory compaction.

4.1. Vibratory compaction modeling

The aforementioned ideal elastoplastic model can describe the behavior of soils during compaction. In this study, the simulant is modeled as an ideal elastoplastic material. According to the ideal elastoplastic theory (as shown in Fig. 7), the motion of the simulant during compaction can be divided into three processes: elastic deformation, plastic deformation, and resilient motion.

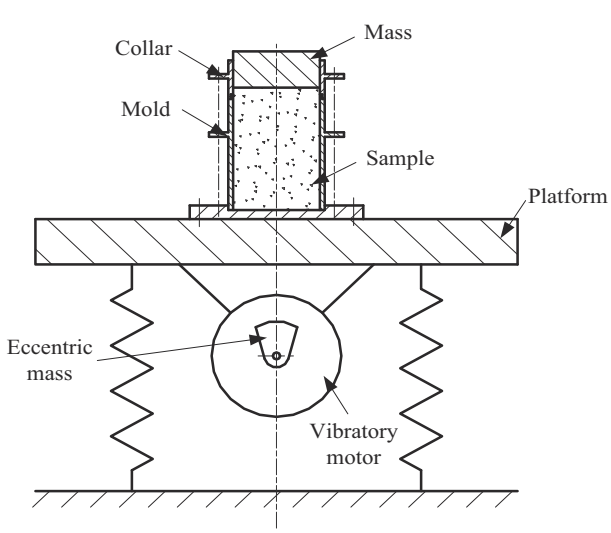

Fig. 6. Vibratory table.

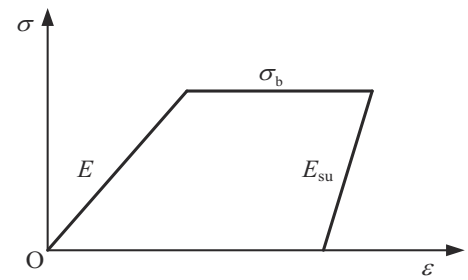

Fig. 7. Stress–strain relation of the ideal elastoplastic model.

The motion equation of the elastic deformation process is expressed as follows:

$$m_p\ddot{x}_p + k_s x_p = 0 \tag{8}$$

where $k_s = 2r_pE/(1-\mu)$ represents the stiffness of the simulant; r_p, the radius of the mass; μ, the Poisson ratio of the simulant; m_p, the mass of the mass; and x_p, the displacement of the mass.

Solving Eq. (8) gives

$$\begin{cases} x_p = & \frac{m_p a_t}{k_s} = \frac{m_p F_0}{m_t k_s}\sin\omega t \\ F_0 = & m_0 e\omega^2 \end{cases} \tag{9}$$

where a_t is the acceleration of the mass; m_t, the mass of the vibratory table; F_0, the impact force; m_0e, the moment of the eccentric mass; and ω, the angular frequency of rotation.

When the simulant is compressed to the plastic limit at $t = t_0$, it undergoes plastic deformation. The motion equation of plastic deformation is expressed as follows:

$$m_p\ddot{x}_p + P_b = 0 \tag{10}$$

where P_b is the plastic limit.

The boundary condition at $t = t_0$ is

$$\begin{cases} x_p(t_0) = & \frac{P_b}{k_s} \\ \dot{x}_p(t_0) = & \frac{\omega\sqrt{m_p^2F_0^2 - P_b m_t^2}}{m_t k_s} \end{cases} \tag{11}$$

Upon substituting Eq. (11) into Eq. (10), the solution of the motion equation of the plastic deformation process can be expressed as follows:

$$x_p = \frac{P_b}{k_s} + \frac{\omega\sqrt{m_p^2F_0^2 - P_b^2m_t^2}}{m_t k_s}(t-t_0) - \frac{P_b}{2m_p}(t-t_0)^2 \tag{12}$$

As the mass reaches the maximum displacement at $t = t_1$, the simulant enters the resilient motion process. The impact force is removed from the simulant, and the elastic strain of the simulant recovers immediately. The equation of the resilient motion is expressed as follows:

$$m_p\ddot{x}_p + k_{su}(x_p\text{-}x_{cr}) = 0 \tag{13}$$

where $k_{su} = 2r_pE_{su}/(1-\mu)$ represents the resilient stiffness of the simulant and x_{cr}, the residual deformation.

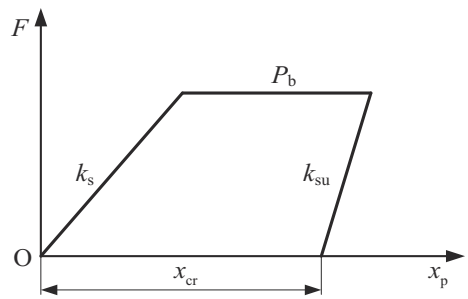

Fig. 8. Relation between simulant deposition and load.

The boundary condition at $t = t_1$ is

$$\begin{cases} x_p(t_1) = & \frac{P_b}{k_s} + \frac{m_p\omega^2}{2k_s^2 P_b}\frac{F_0^2 m_p^4}{m_t^2} - P_b^2 \\ \dot{x}_p(t_1) = & 0 \end{cases} \tag{14}$$

Fig. 8 shows the relation between the load and the displacement of the simulant according to the ideal elastoplastic model.

During vibratory compaction, the plastic deformation of the simulant cannot be recovered, and the elastic deformation of the simulant can be recovered. Upon substituting Eq. (14) into Eq. (13) and according to Fig. 8, the residual deformation can be expressed as follows:

$$x_{cr} = \frac{m_p\omega^2}{2k_s^2 P_b}\left(\frac{F_0^2 m_p^2}{m_t^2} - P_b^2\right) + \frac{P_b}{k_s} - \frac{P_b}{k_{su}} \tag{15}$$

The solution of the equation of the resilient motion is

$$x_p = x_{cr} + \frac{m m_0 e\omega}{m_t k_{su}}\cos\omega(t - t_1) \tag{16}$$

When the impact force is incapable of causing the plastic deformation of the simulant, the elastic deformation of the simulant only occurs during compaction. The elastic deformation of the simulant consumes vibratory energy. Under this condition, the governing equation during compaction can be expressed as follows:

$$x_p = \frac{m_p a_t}{k_s} = \frac{m_p F_0}{m_t k_s}\sin\omega t \tag{17}$$

4.2. Numerical analysis

The weight of the mass, angular frequency of the vibratory table, and eccentric moment and weight of the vibratory table can be determined from the manufacturer's specifications (as shown in Table 3). Two parameters, E and E_{su}, are determined as shown in Section 2. Before preparing the simulant using the vibration compaction method, the simulant was poured into a mold using a paper funnel. The density of the simulant after pouring is 1.8 g/cm^3 as measured by the cutting ring method. The initial density of the simulant during preparation is 1.8 g/cm^3.

Table 3
Parameters of the vibratory table.

Description	Symbol	Value
Eccentric moment (kg·m)	$m_0 e$	0.1
Angular frequency (Hz)	ω	0–50
Mass of vibratory table (kg)	m_t	405.3
Mass of mass (kg)	m_p	3.5
Radius of mass (mm)	r_p	151

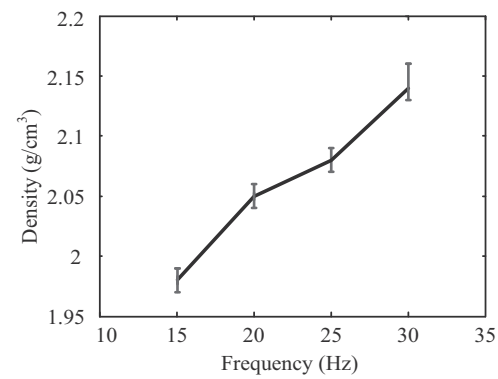

Fig. 9. Relation between FSSD and vibratory frequency.

Previous experimental results indicated that the final steady-state density (FSSD) of the simulant is a function of the vibratory frequency (as shown in Fig. 9). The range bars show the test results in Fig. 9. Eq. (9) gives the relation between the vibratory frequency and the impact force. According to the vibratory compaction model, the plastic limit is proportional to the impact force. Therefore, it can be expressed as a function of the FSSD. According to the Fig. 9 and Eq. (9), the relation between the FSSD and plastic limit of the simulant can be fitted as follows:

$$P_b = 0.07932(100\rho - 184)^2 \tag{18}$$

Numerical simulations of the vibratory compaction model are conducted in MATLAB to analyze the effect of vibratory compaction on the density of the simulant. Fig. 10 shows the density varying time and frequency during compaction. As the vibratory time is consistent, a higher frequency leads to higher density of the simulant. Fig. 10 also shows the trend of higher FSSD for higher frequency. When the simulant is loose, the plastic limit of the simulant is low and the resilient modulus is higher than the compression modulus. As the density of the simulant increases, indicating an increase in the plastic limit and decrease in RCRD, the impact force hardly causes plastic deformation of the simulant. Therefore, vibratory compaction is the most effective when the simulant is loose and the least effective as the density of the simulant approaches 2.14 g/cm^3. When the impact force is incapable of causing the plastic deformation of the simulant, the elastic deformation only occurs in the simulant, and the density of the simulant becomes stable.

5. Experimental verification

This section compares the simulation results of the vibratory compaction model with the experimental results

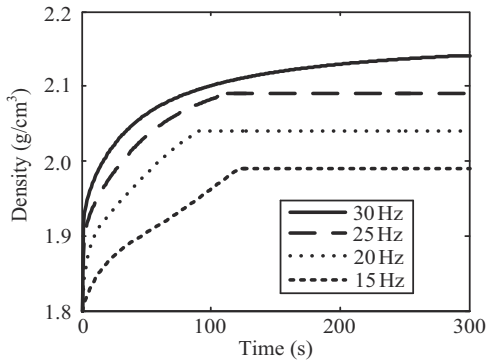

Fig. 10. Simulant density during compaction.

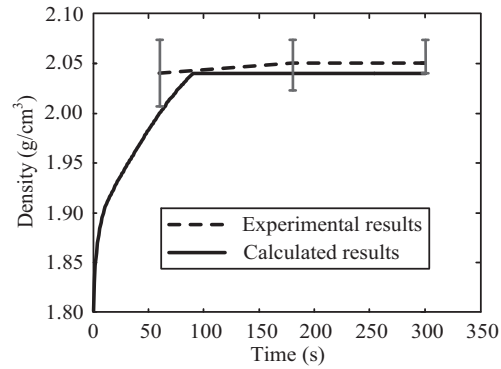

Fig. 12. Comparison of simulant density between the calculated and the experimental results during compaction under vibratory frequency of 20 Hz.

of the vibratory compaction of the simulant. The cutting ring method was employed to test the density of the simulant samples.

In the cutting ring method, a cutting ring is penetrated into a prepared soil sample to acquire a cutting ring sample. As shown in Fig. 11, by using a knife, the surface of the simulant sample is modified to a flat surface. Subsequently, the simulant sample is cut into a column sample whose radius is larger than the inner radius of the cutting ring slightly. The cutting ring is then slowly pushed into the simulant sample. A knife is employed to remove the excess simulant material during the pushing process. A cutting ring sample can be removed after the cutting ring completely penetrates the simulant sample. Subsequently, the excess simulant material placed on both ends of the cutting ring sample is removed using a knife. The modified cutting ring sample is measured by an analytical balance. The weight of the sample is obtained by subtracting the weight of the cutting ring. The density is then calculated by dividing the simulant weight by the inner cutting ring volume. This measurement method has negligible effect on the solid skeleton in a sample, and thus, it causes minimal disruption to the sample.

5.1. Model validation

To validate the influence of the vibratory time on the simulant density, simulant preparation experiments were conducted at a vibratory frequency of 20 Hz for preparation times of 1, 3, and 5 min. As shown in Fig. 12 (the range bars show the test results), the calculated results of the FSSD mainly agree with the measured results. However, the calculation results overestimate the vibratory time by 30 s. As the density obtained is 2.14 g/cm^3, the experimental and calculation results demonstrated that the preparation time should be at least 10 min.

To validate the FSSD predicted by the proposed model, experiments were conducted to prepare simulant samples at vibratory frequencies of 10, 12, 13, 15, 18, 20, 22, 25, 27, and 30 Hz. Each experiment was repeated six times, and it was ensured that the density of the simulant reached its final steady-state value. The experimental results are presented as the mean value of six measured densities. Table 4 shows the FSSD varying vibratory frequencies. The test results demonstrated that the FSSD depends on the vibratory frequency; this was consistent with the prediction of the vibratory compaction model. The FSSD of the calculated results mainly matches the experimental results. Though the calculation results overestimated the vibratory time when loosened simulant samples had to be prepared, a comparison of the experimental results indicated that the proposed model can be used for the preparation of lunar regolith simulant.

The change in simulant density showed unexpected results when the vibratory frequency was lower than 13 Hz. The volume of the simulant remained nearly unchanged during compaction. This situation may be explained by considering the geometrical interlocking of simulant particles. Because the simulant particles are angular and porous, the particle angularity strengthens the interlocking among particles. Even for loose simulant samples, the solid skeleton is more stable than soils with regular particles. Mild impact force hardly destroys and recombines the solid skeleton. Therefore, when using the

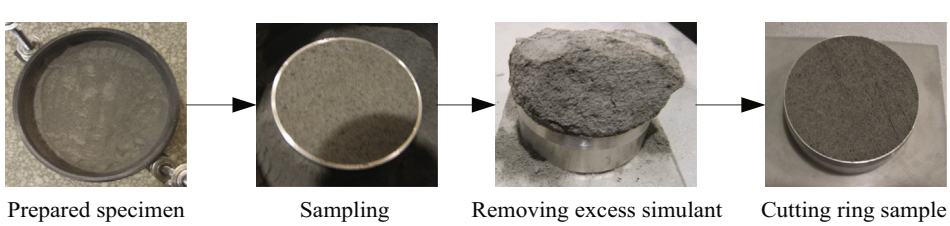

Fig. 11. Cutting ring sampling processes.

Table 4
Final steady-state density of simulant with varying vibratory frequencies.

Vibratory frequency (Hz)	Experimental results (g/cm^3)	Model calculation (g/cm^3)
13	1.96	1.97
15	1.98	1.99
18	2.01	2.02
20	2.05	2.04
22	2.06	2.06
25	2.08	2.09
27	2.11	2.11
30	2.14	2.14

vibratory table presented in this paper to prepare the simulant samples, the appropriate vibratory frequency should be kept higher than 13 Hz.

Both the experimental and the calculated results indicated that an identical effort can result in higher compressibility of a loose simulant sample compared with a dense simulant sample. As a simulant sample is in the loose packing state, the motion of particles is active owing to dynamic loading, and the solid skeleton of the sample can be combined easily. The degree of particle interlocking increases with the density. The solid skeleton is combined as the simulant samples are dense, which results in low compressibility of the dense samples. Therefore, more time is needed for preparing dense simulant samples even though the impact force is powerful. When preparing a sample, the vibratory parameters can be determined by the vibratory compaction model before preparation. The simulant sample can thus be prepared in a repeatable and appropriate manner.

5.2. Longitudinal effects

Previous vibratory compaction experiments for samples with high ratio of depth to radius (RDR) indicated that the simulant density varies along the longitudinal direction when the simulant was prepared by single-layer compaction. The effective range of the impact force is limited along the longitudinal direction, and therefore, the density of the simulant may decrease below a certain depth; this is defined as the longitudinal effect in this study. To investigate the longitudinal effects in vibratory compaction, simulant samples with high RDR were prepared by single-layer vibratory compaction. The cutting ring method was used to investigate the distribution of the simulant density along the longitudinal direction.

The weight of the raw material, sample radius, and frequency of the preparation process are 12.33 kg, 152 mm, and 30 Hz, respectively. To minimize the disruption of the simulant, in each experiment, five or six cutting ring samples were acquired along the longitudinal direction. In this case, cutting ring samples were acquired at a starting point of 20 mm at depths of 20–40, 80–100, 140–160, 200–220, and 260–280 mm. Cutting ring samples were also acquired at a starting point of 0 mm at depths of 0–20, 50–70, 110–130, 170–190, 230–250, and 280–300 mm. Each experiment was repeated nine times, and the results are presented as the mean value of the measured density.

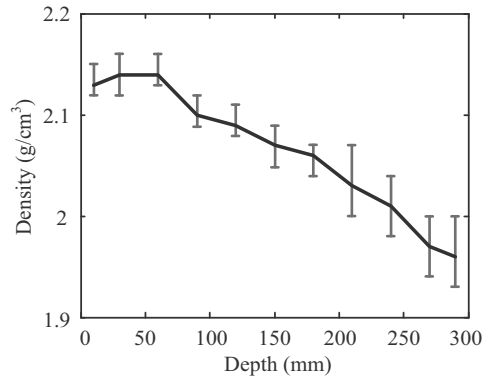

Fig. 13. Simulant density distribution along the longitudinal direction.

As shown in Fig. 13 (the range bars show the test results), a slight fluctuation in density is observed at a depth of $z < 70$ mm. The density decreases considerably in the range 70 mm $< z <$ 100 mm, and it decreases smoothly in the range 100 mm $< z <$ 160 mm. As the depth increases ($z >$ 160 mm), the fluctuation range of the density widens (the fluctuation range of a layer is around 0.08 g/cm^3). The mean value of this density decreases rapidly, and these values become stable at $z >$ 260 mm.

To make the results more general, the parameter z_B (RDR) is used to describe the distribution of the density along the longitudinal direction. According to the density distribution of the simulant sample, the sample is nearly homogenous at $z_B < 0.47$, whereas the influence of the impact force on the simulant decreases significantly at $z_B > 0.47$. This may be influenced by the geometrical interlocking of simulant particles and the friction between the simulant and the mold. Therefore, to prepare a simulant sample in a repeatable and controllable manner, the RDR may be less than 0.47 in single-layer preparation by vibratory compaction. High-RDR samples should be prepared by the multilayer compaction method.

6. Lunar regolith drilling simulant

The aforementioned preparation method can be used to prepare both a homogenous lunar regolith simulant and a simulant whose density decreases along the longitudinal direction with fixed distribution. However, the density of the lunar regolith simulant varies at different landing sites and in different longitudinal directions; it generally increases rapidly in the shallow layer and approaches the maximum value in the deep layer (Mitchell et al., 1974). In the drilling and coring tests, both the density and the shear strength of the lunar regolith drilling simulant are determined. To mimic subsurface lunar regolith, this section presents a modified vibratory-compaction-based preparation method to prepare the lunar regolith drilling simulant.

Fig. 14 summarizes the preparation method of the lunar regolith drilling simulant. To simulate the mechanical property distribution along the longitudinal direction, the simulant sample is prepared according to the following steps: (1) convert the expected mechanical property to the simulant density for preparation, (2) select the vibratory compaction parameters and preparation layers, and (3) prepare the drilling simulant by using the vibratory compaction method. The relation between the shear strength and the density can be estimated by the results of direct shear tests. The preparation layers are determined by the density distribution. As homogenous samples can be obtained in a repeatable manner by single-layer compaction, the drilling simulant must be divided into several layers to approach the distribution before preparation. More preparation layers will provide higher approximation accuracy but need more preparation time. The vibratory parameters are determined by the density of each layer according to the vibratory compaction model.

A drilling simulant sample whose density varies along the longitudinal direction was prepared to validate the modified preparation method. The preparation target was the density distribution given by Eq. 1 with $\rho_0 = 1.92$ and $k = 0.07$. To mimic this distribution, the sample was obtained from five-layer simulant vibratory compaction, and the density of each layer was estimated by Eq. 1. The vibratory parameters of each layer can be determined by the vibratory compaction model. Fig. 15 shows a comparison between the expected and the experimental results. The experimental results are mainly consistent with the expected results at the measurement point. However, the drilling simulant is homogenous in each layer, and the variation range of the density is less than expected. To approach the desired density distribution, the sample should be divided into more layers before preparation. The shear strength of the drilling simulant can be acquired by substituting the density of the simulant into Eq. 7. The shear strength of the simulant (see Table 5) is within the range of the shear strength of actual lunar regolith.

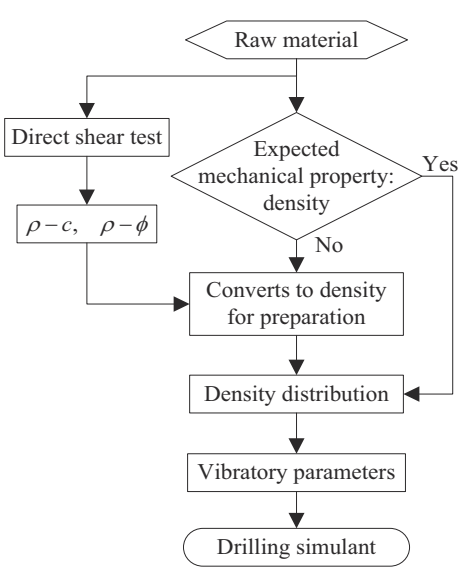

Fig. 14. Flowchart of drilling simulant preparation.

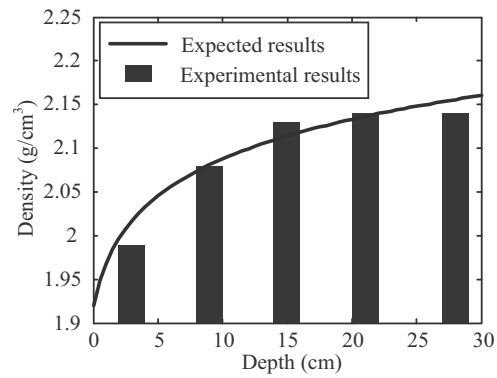

Fig. 15. Density distribution of a drilling simulant sample.

Table 5
Density and shear strength of the simulant sample

Tested depth (mm)	Tested density (g/cm^3)	ϕ (°)	c (kPa)
20–40	1.99	29.7	0.66
80–100	2.08	33.1	1.63
140–160	2.13	34.7	2.61
200–220	2.14	35.1	2.85
270–290	2.14	35.1	2.85

In the aforementioned preparation processes, the density was chosen as the simulation target. If the shear strength is chosen as the simulation target, the corresponding simulant density can be obtained by substituting the friction angle or cohesion of the simulant into Eq. (7). Subsequently, the sample can be obtained by using the preparation method presented in this paper.

7. Conclusions

This paper presents a vibratory-compaction-based method to prepare lunar regolith drilling simulant with mechanical properties varying along the longitudinal direction. A mathematical model based on the ideal elastoplastic theory was developed to describe the interaction between the mass and the lunar regolith simulant. The model captures the relation between the vibratory parameters and the regolith simulant density. A comparison of the vibratory compaction experiments and the calculated results demonstrated that the vibratory compaction model can effectively predict the FSSD during vibratory compaction. The model provides a guide for regolith simulant preparation using the vibratory compaction method. Considering the heterogeneity of the simulant samples, experiments were conducted to analyze the density distribution along the longitudinal direction for high-RDR simulant samples. An appropriate range of RDR was presented to facilitate simulant preparation. Inspired by the Apollo shear strength analysis method, a method was presented to build a relation between the preparation parameters and the expected mechanical property of the drilling simulant. By

using the multilayer preparation method presented in this paper, lunar regolith drilling simulant with desired mechanical property distribution can be acquired to mimic actual lunar regolith along the depth direction. The lunar regolith is dense and yet, it is loose. The vibration compaction method can only be used to prepare simulant samples with medium or high density. If a loose drilling simulant has to be prepared, other preparation methods such as the pour and rain methods should be developed in future works.

Acknowledgements

The authors thank the supports by National Natural Science Foundation of China (Nos. 61403106, 51575122), Program of Introducing Talents of Discipline to Universities (No. B07018), and Fundamental Research Founds for the Central Universities (No. HIT.NSRIF.2014051).

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